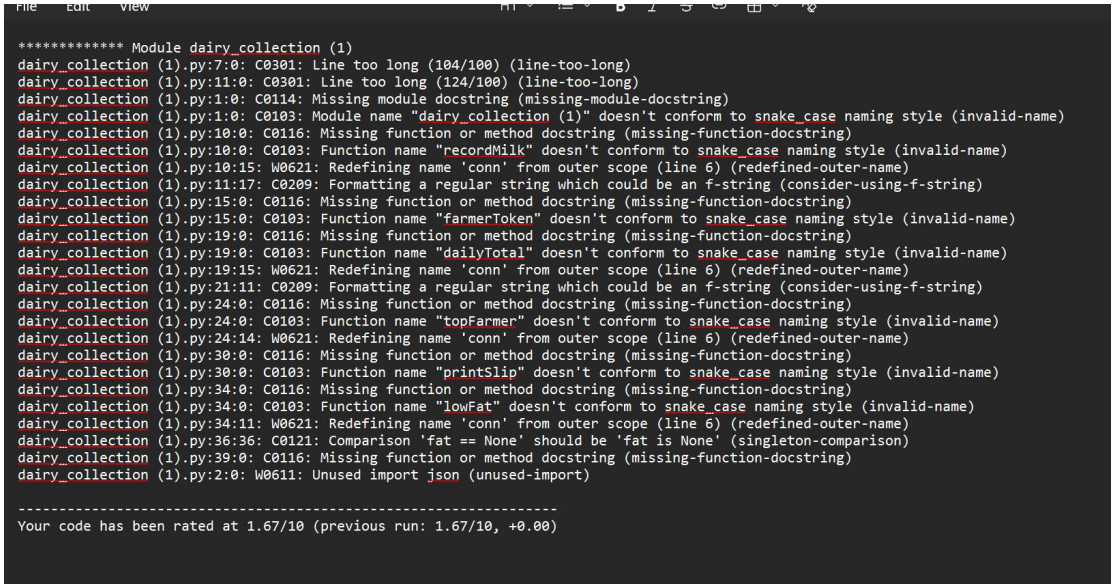
SOFTWARE ENGINNERING WEEK - 5 LAB

NAME :SHASHWATH K R
SRN:PES2UG24CS463
SECTION:H

BEFORE CHANGE IN Pylint



```
File   Edit   View

************* Module dairy_collection (1)
dairy_collection (1).py:7:0: C0301: Line too long (104/100) (line-too-long)
dairy_collection (1).py:11:0: C0301: Line too long (124/100) (line-too-long)
dairy_collection (1).py:1:0: C0114: Missing module docstring (missing-module-docstring)
dairy_collection (1).py:1:0: C0103: Module name "dairy_collection (1)" doesn't conform to snake_case naming style (invalid-name)
dairy_collection (1).py:10:0: C0116: Missing function or method docstring (missing-function-docstring)
dairy_collection (1).py:10:0: C0103: Function name "recordMilk" doesn't conform to snake_case naming style (invalid-name)
dairy_collection (1).py:10:15: W0621: Redefining name 'conn' from outer scope (line 6) (redefined-outer-name)
dairy_collection (1).py:11:17: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
dairy_collection (1).py:15:0: C0116: Missing function or method docstring (missing-function-docstring)
dairy_collection (1).py:15:0: C0103: Function name "farmerToken" doesn't conform to snake_case naming style (invalid-name)
dairy_collection (1).py:19:0: C0116: Missing function or method docstring (missing-function-docstring)
dairy_collection (1).py:19:0: C0103: Function name "dailyTotal" doesn't conform to snake_case naming style (invalid-name)
dairy_collection (1).py:19:15: W0621: Redefining name 'conn' from outer scope (line 6) (redefined-outer-name)
dairy_collection (1).py:21:11: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
dairy_collection (1).py:24:0: C0116: Missing function or method docstring (missing-function-docstring)
dairy_collection (1).py:24:0: C0103: Function name "topFarmer" doesn't conform to snake_case naming style (invalid-name)
dairy_collection (1).py:24:14: W0621: Redefining name 'conn' from outer scope (line 6) (redefined-outer-name)
dairy_collection (1).py:30:0: C0116: Missing function or method docstring (missing-function-docstring)
dairy_collection (1).py:30:0: C0103: Function name "printSlip" doesn't conform to snake_case naming style (invalid-name)
dairy_collection (1).py:34:0: C0116: Missing function or method docstring (missing-function-docstring)
dairy_collection (1).py:34:0: C0103: Function name "lowFat" doesn't conform to snake_case naming style (invalid-name)
dairy_collection (1).py:34:11: W0621: Redefining name 'conn' from outer scope (line 6) (redefined-outer-name)
dairy_collection (1).py:36:36: C0121: Comparison 'fat == None' should be 'fat is None' (singleton-comparison)
dairy_collection (1).py:39:0: C0116: Missing function or method docstring (missing-function-docstring)
dairy_collection (1).py:2:0: W0611: Unused import json (unused-import)

-------------------------------------------------------------------
Your code has been rated at 1.67/10 (previous run: 1.67/10, +0.00)
```

BEFORE CHANGE IN Bandit

```
Run started:2026-10-14 06:46:13.101257+00:00

Test results:
>> Issue: [B608:hardcoded_sql_expressions] Possible SQL injection vector through string-based query construction.
   Severity: Medium   Confidence: Medium
   CWE: CWE-89 (https://cwe.mitre.org/data/definitions/89.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b608_hardcoded_sql_expressions.html
   Location: .\dairy_collection (1).py:11:17
10      def recordMilk(conn, farmer, litres, fat, day):
11          conn.execute("INSERT INTO milk (farmer, litres, fat, day) VALUES ('{}', {}, {}, '{}')".format(farmer, litres, fat, day))
12          conn.commit()

--------------------------------------------------
>> Issue: [B324:hashlib] Use of weak MD5 hash for security. Consider usedforsecurity=False
   Severity: High   Confidence: High
   CWE: CWE-327 (https://cwe.mitre.org/data/definitions/327.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b324_hashlib.html
   Location: .\dairy_collection (1).py:16:11
15      def farmerToken(farmer):
16          return hashlib.md5(farmer.encode()).hexdigest()
17

--------------------------------------------------
>> Issue: [B605:start_process_with_a_shell] Starting a process with a shell, possible injection detected, security issue.
   Severity: High   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b605_start_process_with_a_shell.html
   Location: .\dairy_collection (1).py:31:4
30      def printSlip(farmer):
31          os.system("echo printing slip for " + farmer)
32

--------------------------------------------------

Code scanned:
```

```
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b324_hashlib.html
   Location: .\dairy_collection (1).py:16:11
15      def farmerToken(farmer):
16          return hashlib.md5(farmer.encode()).hexdigest()
17

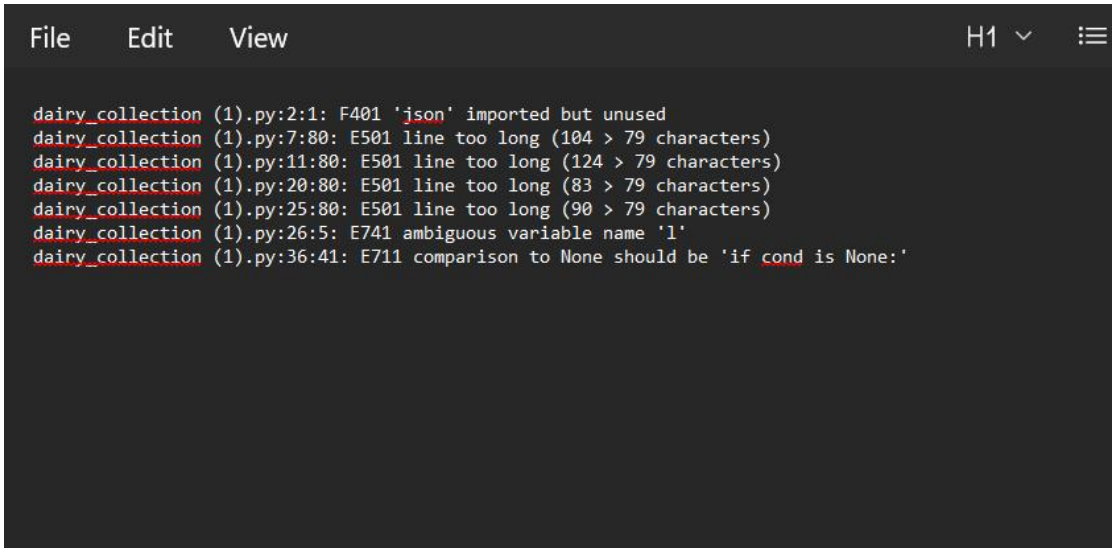
--------------------------------------------------
>> Issue: [B605:start_process_with_a_shell] Starting a process with a shell, possible injection detected, security issue.
   Severity: High   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b605_start_process_with_a_shell.html
   Location: .\dairy_collection (1).py:31:4
30      def printSlip(farmer):
31          os.system("echo printing slip for " + farmer)
32

--------------------------------------------------

Code scanned:
        Total lines of code: 30
        Total lines skipped (#nosec): 0
        Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

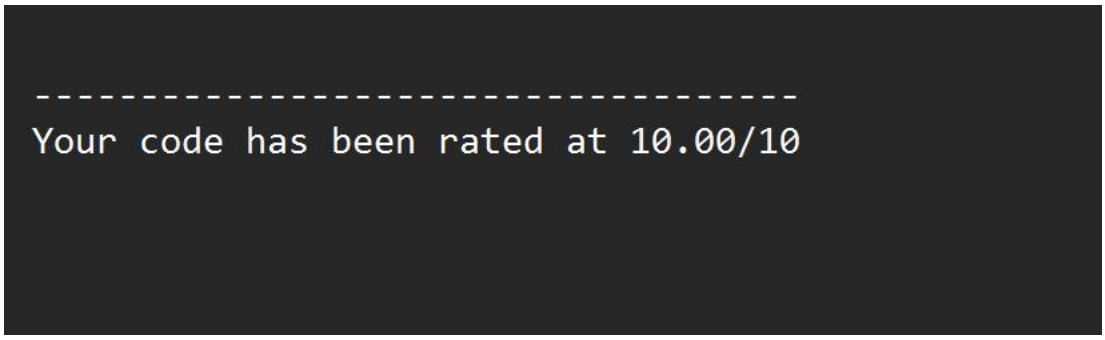
Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 0
                Medium: 1
                High: 2
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 1
                High: 2
Files skipped (0):
```

BEFORE CHANGE IN Flake8



AFTER CHANGING CODE

AFTER CHANGE IN Pylint



AFTER CHANGE IN Bandit

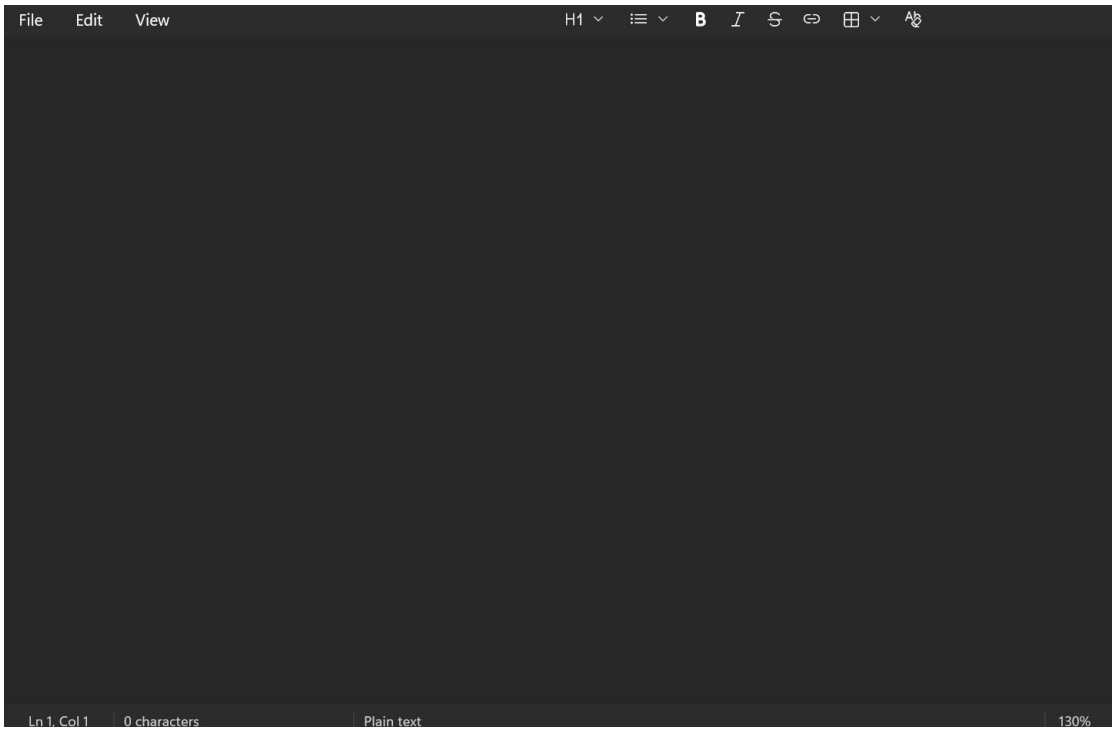
```
Run started:2026-10-14 09:06:00.081709+00:00

Test results:
        No issues identified.

Code scanned:
        Total lines of code: 83
        Total lines skipped (#nosec): 0
        Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
Files skipped (0):
```

AFTER CHANGE IN Flake8



CODE:

```
"""Track dairy farmers' milk collections using an in-memory
SQLite database."""

import hashlib
```

```
import sqlite3
```

```
def create_connection() -> sqlite3.Connection:
    """Create an in-memory database and its milk collection
table."""
    connection = sqlite3.connect(":memory:")
    connection.execute(
        """
        CREATE TABLE milk (
            id INTEGER PRIMARY KEY,
            farmer TEXT NOT NULL,
            litres REAL NOT NULL,
            fat REAL,
            day TEXT NOT NULL
        )
        """
    )
    return connection
```

```
def record_milk(
    connection: sqlite3.Connection,
    farmer: str,
    litres: float,
    fat: float,
    day: str,
) -> None:
    """Record one farmer's milk collection using a
parameterized query."""
    connection.execute(
        """
        INSERT INTO milk (farmer, litres, fat, day)
        VALUES (?, ?, ?, ?)
        """,
        (farmer, litres, fat, day),
    )
    connection.commit()
```

```
def farmer_token(farmer: str) -> str:
    """Return a SHA-256 hexadecimal digest for a farmer's
name."""
    return hashlib.sha256(farmer.encode("utf-8")).hexdigest()
```

```
def daily_total(connection: sqlite3.Connection, day: str) ->
str:
```

```
    """Return the total litres of milk collected on a given
day."""
    row = connection.execute(
        "SELECT COALESCE(SUM(litres), 0) FROM milk WHERE day
= ?",
        (day,),
    ).fetchone()
    total_litres = row[0]
    return f"Total: {total_litres:.1f} litres"
```

```
def top_farmer(connection: sqlite3.Connection) -> str | None:
    """Return the farmer with the highest total milk volume,
if available."""
    row = connection.execute(
        """
        SELECT farmer, SUM(litres) AS total_litres
        FROM milk
        GROUP BY farmer
        ORDER BY total_litres DESC, farmer ASC
        LIMIT 1
        """
    ).fetchone()
    return row[0] if row else None
```

```
def print_slip(farmer: str) -> None:
    """Print a collection-slip message without invoking a
shell."""
    print(f"Printing slip for {farmer}")
```

```
def low_fat(connection: sqlite3.Connection) -> list[str]:
    """Return unique farmers with missing fat values or fat
below 3.5."""
    rows = connection.execute(
        """
        SELECT DISTINCT farmer
        FROM milk
        WHERE fat IS NULL OR fat < ?
        ORDER BY farmer
        """,
        (3.5,),
    ).fetchall()
    return [row[0] for row in rows]
```

```
def main() -> None:
```

```
    """Run a demonstration of the dairy milk collection
tracker."""
    connection = create_connection()
    try:
        record_milk(connection, "Gopal", 12.5, 4.1, "Mon")
        record_milk(connection, "Lata", 20.0, 3.2, "Mon")
```

```
        print(daily_total(connection, "Mon"))
        print("Top farmer:", top_farmer(connection))
        print("Low-fat farmers:", low_fat(connection))
        print_slip("Gopal")
        print("Farmer token:", farmer_token("Lata"))
    finally:
        connection.close()
```

```
if __name__ == "__main__":
    main()
```

Before and After Code Analysis

No	Tool	Before	After	Status
1	Pylint	Score: 1.67/10	Score: 10.00/10	Passed — above 9.5
2	Flake8	Multiple formatting violations	No violations	Passed
3	Bandit	3 security findings	No issues identified	Passed